A Synchronized Growing-Prefix Bridge for Noisy Traces and Monodromy Counterloops

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Abstract

The monodromy counterloop bridge was previously coefficientwise: every fixed order converges, but no order window was allowed to grow with vanishing noise. We prove a diagonal synchronization theorem. There exist a prefix clock $h_\sigma \to \infty$ and finite-cycle clock $k_\sigma \to \infty$, with $h_\sigma < 2k_\sigma$, such that the actual noisy parity-extracted trace minus the finite-radius counterloop moment converges uniformly to the deterministic numerator anchor over all orders $2 \le n \le h_\sigma$. The theorem is exact and uses the fixed-order noisy trace limit together with the RH-17 radius law. It is deliberately nonquantitative. An unweighted growing prefix does not control a determinant on a disk of radius greater than one, and no noisy spectral-submultiset identification follows.

1 Inputs

Let $c_{\sigma,n}$ be the Hardy-scaled noisy trace after the two peripheral branches are removed. RH-11, RH-14, and RH-15 give, for each fixed $n \ge 2$,

$$c_{\sigma,n} \longrightarrow c_n.$$

Let $s_{k,n}$ be the RH-272 finite-radius counterloop moment. For fixed n and sufficiently large k, there is no alias and

$$s_{k,n} \longrightarrow p_n, \qquad a_n = c_n - p_n,$$

where a_n is the deterministic numerator coefficient anchor.

2 Diagonal theorem

Theorem 2.1. *There exist integer-valued functions h_σ, k_σ such that*

$$h_\sigma \to \infty, \qquad k_\sigma \to \infty, \qquad h_\sigma < 2k_\sigma,$$

and

$$\max_{2 \le n \le h_\sigma} |c_{\sigma,n} - s_{k_\sigma,n} - a_n| \longrightarrow 0. \tag{1}$$

Proof. For every integer $j \ge 2$, fixed-order convergence supplies $\delta_j > 0$ such that

$$\max_{2 \le n \le j} |c_{\sigma,n} - c_n| \le j^{-1} \quad (0 < \sigma \le \delta_j).$$

Choose the δ_j strictly decreasing with $\delta_j \le j^{-1}$. The finite-radius moment law supplies a threshold K_j^0 such that, for every $k \ge K_j^0$,

$$\max_{2 \le n \le j} |s_{k,n} - p_n| \le j^{-1}.$$

With $K_1 = 0$, choose recursively $K_j > \max\{K_{j-1}, j/2, K_j^0\}$. On the slab $\delta_{j+1} < \sigma \le \delta_j$, set $h_\sigma = j$ and $k_\sigma = K_j$. Then the triangle inequality gives a maximum in (1) no larger than $2/j$, while both clocks diverge. For $\sigma > \delta_2$, define the two clocks arbitrarily. $\square$

3 What the theorem does not weight

For $R > 1$, a bound $\max_{n \le h} |e_n| \le \varepsilon_h$ gives only

$$\sum_{n=2}^{h} \frac{|e_n| R^n}{n} \le \varepsilon_h \sum_{n=2}^{h} \frac{R^n}{n}.$$

The geometric sum may grow like R^h/h. The diagonal construction does not force $\varepsilon_h R^h \to 0$. A separate weighted-prefix estimate is therefore necessary before analytic determinant gluing.

Remark 3.1. The result is an actual-noisy trace theorem, not a noisy-root theorem. It does not identify counterloop atoms as an algebraic spectral submultiset and does not close Gates A–E or imply any Hilbert–Polya or RH statement.